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Mapping J1939 Signals to Vehicle Application Subsystems

DOCUMENT APPROVAL

	Author	Reviewer	Approver
Name	Paul Shipley		
Role	Software Engineer		
Date	2015-01-27		
Comments			

DOCUMENT INFORMATION

Document Reference	
Document Status	Draft
Author	Paul Shipley
Revision	2.0
Issue Date	23rd August 2021
Archive Location	

REVISION HISTORY

Revision	Date	Details	Author	Change Number
1.0	27 Jan 2015	Initial version	PS	
2.0	23 Aug 2021	Updated template	DH	

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1 Introduction

1.1 Purpose and Scope

Explain the I-protocol mapping to Vehicle App (TracApp, PumpApp and BattApp).

1.2 Related documents

Ref	Doc Ref	Title	Origin	Revision
1				
2				

Table 1. Related Documents

1.3 Document Conventions and Terminology

Typeface	Indicates
 **WARNING**	Very important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent serious problems.
 CAUTION	Important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent undesirable problems.
 *NOTE*	Useful information for the document user.

Table 2. Typographical Conventions

1.4 Glossary

Term	Description

Table 3. Glossary

1.5 Conditions for document usage

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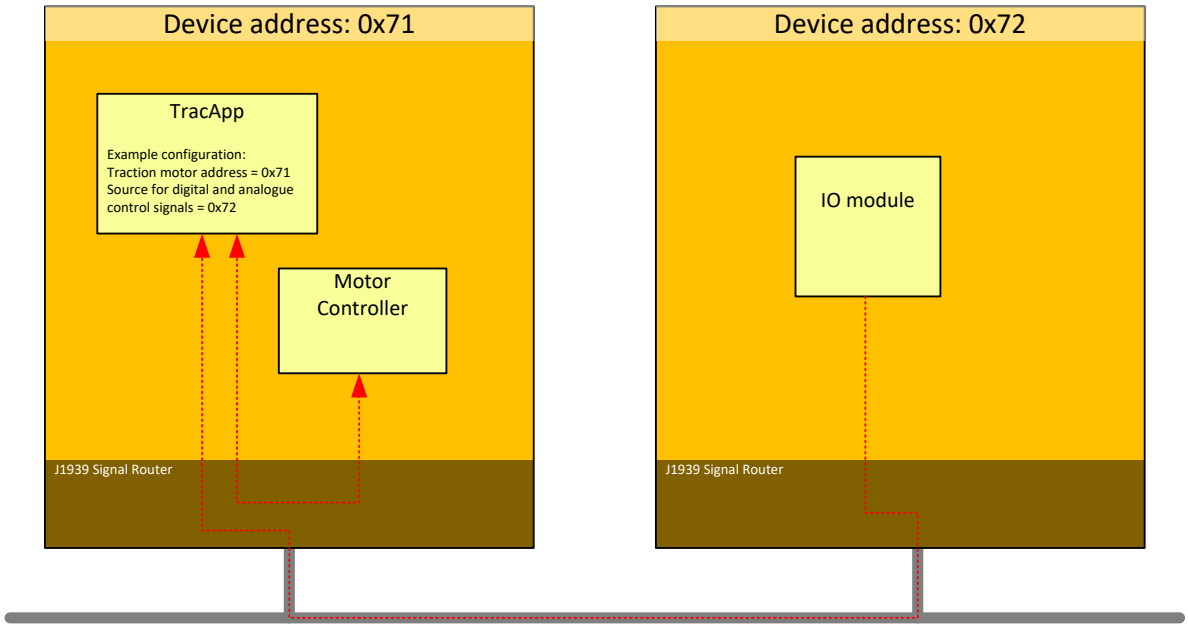
2 Introduction

2.1 Overview of Routing System

The J1939 message routing sub-system is included in Genus (HVLP, D8, Gen5) builds as a means to map H-protocol and I-protocol signals between devices. Both of these protocols use J1939 style messaging formats and, since J1939 provides its own mechanism for providing peer-to-peer communication, it is sensible to include this as part of our approach to signal mapping.

The approach to signal routing is actually very simple. In fact, the source address and destination address for all messages of this type are encoded into the message IDs (see section 5).

The J1939 routing mechanism also lends itself well to modular software architecture. Signals can be sent to motor controllers and IO modules anywhere in the system. The protocol takes care of accessing the information and ensuring it reaches the correct destination, regardless of whether the module that any sub-system wishes to communicate with is on a local or a remote node.



In an example system, we can say to a TracApp module running anywhere in the system, the motor you are controlling is on the device with address 0x71, and control signals will be acquired from the device with address 0x72. The J1939 signal routing system will then set up communication paths between these blocks, using CAN where necessary. If CAN is not required, the signals will be routed internally automatically.

2.2 Departure from PDOs

The J1939 mechanism is a radical departure from the PDO mechanism used by CANopen. Process data in Genus is no longer carried using the PDO protocol, but is handled completely by the J1939 router. You may notice the following differences in the object dictionary:

- VPDO configuration objects in the range 0x3000 to 0x3400 no longer exist. If signals are to be configured to be on the local node, they can be set up in exactly the same way as if they were on a remote node, as in the example given above for motor signals.
- Previously, a number of objects existed in the 0x2000 to 0x26FF range. These objects represented vehicle functions, and were intended to be connected to PDO maps to define various system maps. These have been removed, as signal routing is no longer handled by PDO.

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Whereas previously the PDO mechanism allowed the user to specify any COB-ID for a set of signals and map them as required, the system was difficult to configure and prone to error. The H-protocol and I-protocol now set out specific message layouts, and the routing mechanism now automatically calculates CAN message identifiers. While this might not be as flexible as the PDO system, it becomes very simple to configure.

The standard TPDO mechanism may still be used for debugging with the DVT's vehicle interface viewer if desired.

2.3 DVT helper Input/Output window

The recommended method for configuring the J1939 mapping is to use the Input / Output tab in the DVT helper. If the connected node (or dcf being edited) supports J1939 I-protocol mapping then this configuration is shown instead of the VPDO mapping used in Gen4.

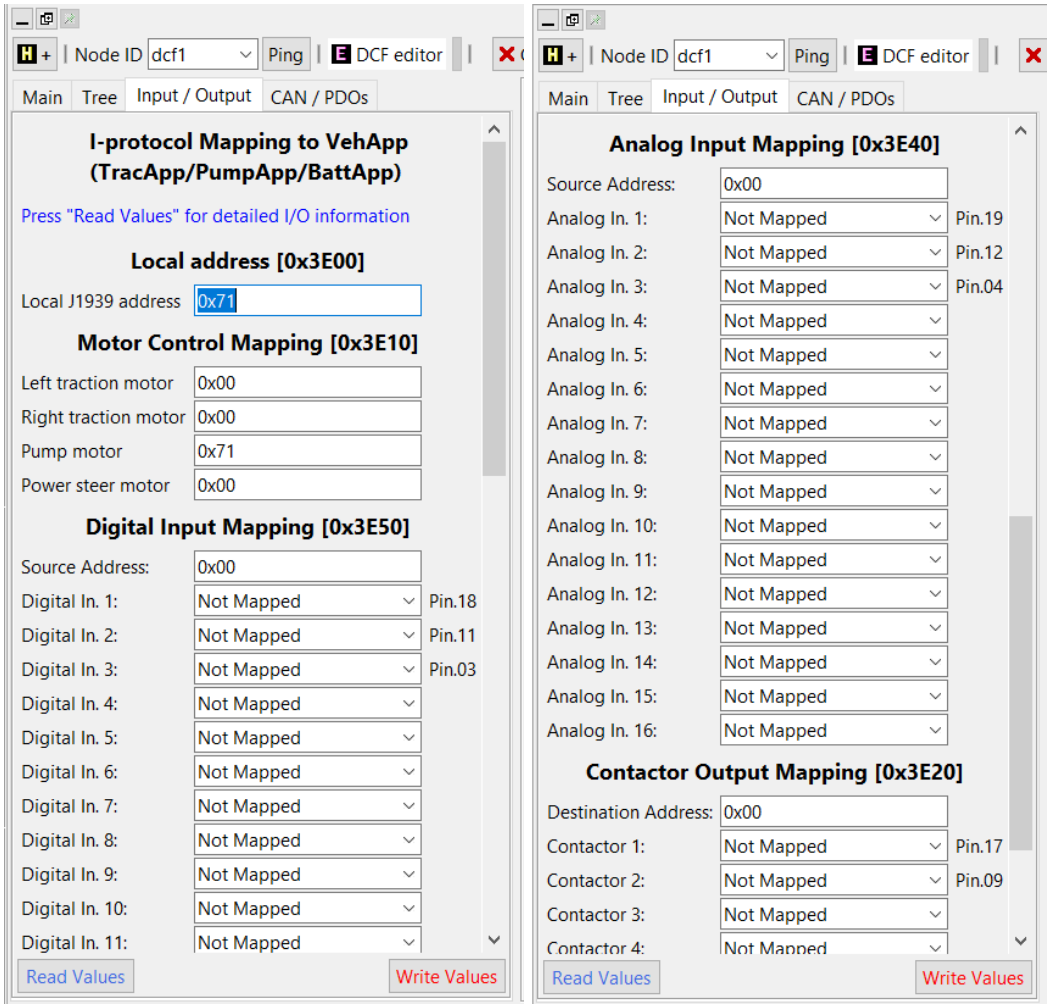


Figure 1 - DVT helper

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3 Configuration

The basic J1939 parameters at object 0x3E00 need to be set on all nodes.

The mapping parameters from objects 0x3E10 to 0x3E5F only need to be set on the node acting as the vehicle master.

3.1 Basic J1939 Configuration

Basic J1939 parameters must be set at object 0x3E00. Principally, this consists of the device's address, which is set at object 0x3E00,1. This must be unique throughout the system. (Note: future implementations of software are expected to include the J1939 address claim mechanisms to ensure that this value is unique, and raise an error if this is not the case).

The J1939 address remains separate to the CANopen node ID. Do not confuse the two – they are totally individual parameters. The CANopen node ID remains in place for configuration using the existing SDO protocol, while the J1939 address is used for signal routing.

Be aware of the terms "node ID" and "address" that are used throughout this document, which refer to and distinguish between the CANopen and J1939 parts respectively.

The other thing that needs to be set at object 0x3E00 is sub-index 2. If this is set to 0x01, the message routing system will copy all data to CAN regardless of whether the data is required remotely or not. This is useful for debugging over CAN, as it allows all signals to be monitored. If this is set to 0x00 then only signals that need to go over CAN will be transmitted, which can be used to cut down network traffic on a busy CANbus, once the system configuration has been validated.

3.2 Motor Mapping

The vehicle application subsystems have the ability to control four motors:

- A left traction motor
- A right traction motor
- A pump motor
- A power steer motor

Signals for any of these motors can be routed over the CANbus. All that needs to be set is the source address of the device that is controlling the respective motor.

For example, say we have a system with a single traction motor and a pump motor. The traction motor is connected to the controller with source address 0x72, and the pump motor is connected to the controller with source address 0x73. Object 0x3E10, which is the object that sets up motor control mappings, would be set as follows:

```
0x3E10, 1 – 0x72
0x3E10, 2 – 0x00
0x3E10, 3 – 0x73
0x3E10, 4 – 0x00
```

For motors that aren't present in the system, set the address to zero.

Once this is done, the signal mapping system will automatically configure the H-protocol messages such that control data for the motors is sent to the correct device, and status data is received from it.

3.3 Signal Mapping

IO signal mapping uses I-protocol messaging. This is configured in the object dictionary at objects in the range 0x3E20 to 0x3E5F.

All analogue and digital inputs are configured in the same way. Each set of signals comprises of an object where sub-index 1 is intended to be set to the J1939 source address of the device that is to send or receive the signal. The remaining sub-indices should be set to the handle values of the functions that are attached to that device.

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3.3.1 Example Digital Input Mapping

For example, say we have a number of digital inputs connected to a device that supports I-protocol, and is configured with source address 0x71. We want digital input 1 to be a forward switch, digital input 2 to be a reverse switch, and digital input 3 to be FS1. Digital inputs are configured by objects in the range 0x3E50 to 0x3E5F, but for this example we'll use the first digital input set at 0x3E50. In this case, object 0x3E50 would be set as follows:

```
0x3E50, 1 – 0x71
0x3E50, 2 – 0x0201
0x3E50, 3 – 0x0202
0x3E50, 4 – 0x0203
0x3E50, 5 – 0x0000
0x3E50, 6 – 0x0000
0x3E50, 7 – 0x0000
0x3E50, 8 – 0x0000
0x3E50, 9 – 0x0000
0x3E50, 10 – 0x0000
0x3E50, 11 – 0x0000
0x3E50, 12 – 0x0000
0x3E50, 13 – 0x0000
0x3E50, 14 – 0x0000
0x3E50, 15 – 0x0000
0x3E50, 16 – 0x0000
0x3E50, 17 - 0x0000
```

To recap:

- Sub index 1 in the above example is set to 0x71, because this is the source address of the device to which these digital inputs are configured.
- There are 16 digital inputs on devices that support I-protocol. The functions of each of the 16 digital inputs on source address 0x71 is configured at sub-indices 2 to 17.
- The values 0x0201, 0x0202 and 0x0203 are the handle values for the forward, reverse and FS1 switch inputs. These values are determined by examining the lookup tables in section 6.
- Anything that isn't configured can be left set to 0x0000.

In this example, the digital inputs are connected to a device with source address 0x71. For the purposes of configuration, it doesn't matter which device has been configured with source address 0x71. It could be the same device that is running the vehicle application subsystems, in which case the signals will be routed internally through the controller's virtual CANbus. If source address 0x71 is a remote node, the J1939 message routing mechanism is used. The controller will automatically detect and use CAN messages that have been sent with a J1939 source address of 0x71.

3.3.2 Example Analogue Output Mapping

For analogue outputs, we follow a similar pattern to the digital inputs. Again, I-protocol is used to carry the signal information, only the flow of information is in the opposite direction.

For example, say our device at source address 0x71 has contactor drive 1 wired to the line contactor, and contactor drive 2 wired to an external LED. Configuration of analogue outputs is handled by objects in the range 0x3E20 to 0x3E2F, but for this example we'll use 0x3E20:

```
0x3E20, 1 – 0x71
0x3E20, 2 – 0x0500
0x3E20, 3 – 0x0504
0x3E20, 4 – 0x0000
0x3E20, 5 – 0x0000
0x3E20, 6 – 0x0000
0x3E20, 7 – 0x0000
0x3E20, 8 – 0x0000
0x3E20, 9 – 0x0000
```

To recap:

- Sub index 1 is set to 0x71 because that is the destination address that we want the data to go to.
- I-protocol supports 8 contactor drives per destination address
- The values 0x0500 and 0x0504 are the handle values for the line contactor and external LED respectively. These values are determined by looking at the tables in section 6.

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3.4 Signal Mapping Verification

On the master node, object 0x3E01 can be used to check that IO signals are being received correctly, and are attached to the correct physical input.

To use this object, first set up the mappings as described above. The system should be operational so that signal process data is being transmitted and received by all nodes.

Now, write the handle of the signal that you wish to test into 0x3E01,1. The handle for any function can be found in the tables in section 6. For example, if we want to test the forward switch, write a value of 0x0201 to object 0x3E01,1.

Finally, read back object 0x3E01,2. The value of this object should reflect the state of the forward switch. When toggling the physical forward switch input, you should be able to see a response when reading back object 0x3E01,2.

This technique works for all IO signals, both digital and analogue. For instance, if you want to check the throttle voltage, set object 0x3E01,1 to 0x0300. If the system is configured correctly you should now get a throttle voltage signal from object 0x3E01,2. This trick is especially useful for configuring the zero push and full push voltages in the throttle configuration object, 0x2910.

Outputs can also be monitored using this technique. For example, if we write 0x0504, which is the handle for the external LED, to object 0x3E01,1, we should be able to observe the LED state at 0x3E01,2.

Four sets of debug items are provided in 0x3E01, allowing four signals to be monitored simultaneously. Note, these are provided for debug only, and the values written to this object are not stored in NVM.

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4 Required Configuration

Some additional items need to be configured within the H-protocol and I-protocol objects in order to be compatible with the vehicle application subsystem. These are noted here for convenience.

4.1 H-Protocol

For H-protocol, some timeout values need to be set to suit the data rates expected by TracApp. Since TracApp only produces data with a 20ms period, the timeout in the motor control side needs to be extended. Normally this would be three times the period that is normally expected. Also, the transmit rate can be divided by a factor of four, since data will only be processed by TracApp at a 20ms rate.

In short, the following values can be set:

```
0x4655,3 = 0x0C
0x4655,4 = 0x04
```

If you want to see motor control debug data, set 0x4655,5 to be 0x04. Otherwise, set it to zero.

The vehicle application subsystem does not yet generate checksum and counter data. The check in the motor control needs to be disabled by setting 0x4655,2 to 0xE38E.

Object 0x4655,1 should be set to either 0x5533 to have the motor drive normally, or 0xCCAA to invert the direction of forward rotation.

4.2 I-Protocol

The following objects should be set to configure the default PGNs for I-protocol, with the default transmission rates:

```
0x3F01,1 = 50
0x3F01,2 = 50
0x3F01,3 = 0
0x3F01,4 = 6
0x3F01,5 = 1
0x3F01,7 = 0x40
0x3F01,8 = 0x41
0x3F01,9 = 0x48
0x3F01,10 = 0x49
0x3F01,11 = 0x4A
0x3F01,12 = 0x4B
0x3F01,13 = 0x4C
```

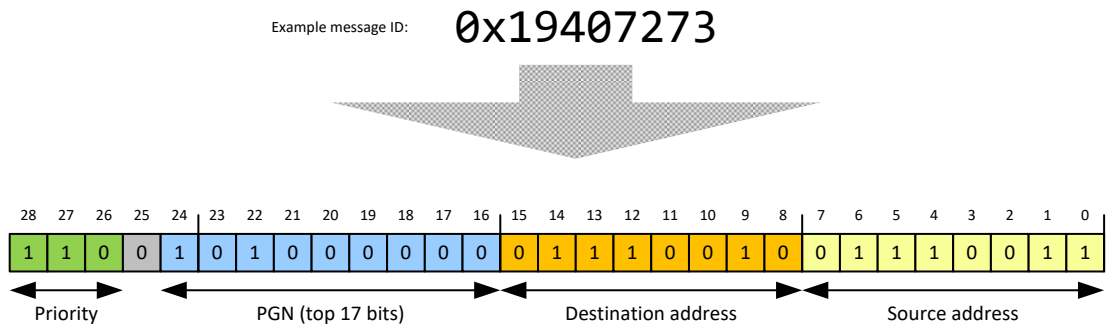
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5 J1939 Signal Messaging Detail

This section explains how J1939 actually manages to route messages between devices. It is probably of most interest to engineers during the system integration stage, attempting to verify or debug signal messaging.

5.1 Example Signal

Consider the bit layout for a 29 bit message ID, encoded using J1939's PDU1 format[1], for the following example message ID:



For these signals, it is easy to see the source address of the data, and its intended destination address. Generally speaking, for H-protocol and I-protocol control and status information, the source address and destination address are both encoded into the bottom 16-bits of the message ID. In this case, the address of the device that has produced this message is 0x73, and the message is intended for the device that has address 0x72.

Sometimes the destination address appears as 0xFF. This is ok; it just means that the data isn't intended for any specific device. For both H-protocol and I-protocol, status information is transmitted with a destination address of 0xFF. This means it isn't intended for any specific device address, and can in fact be picked up by several devices. This includes display, traction controllers, or logging tools. However, the destination address cannot be the broadcast address for control information. In this case, the vehicle application sub-systems must specify which address the control data is intended for.

The source address will (nearly[2]) always be populated.

The PGN part of the message ID specifies what the content of the message is. In this case, the PGN is 0x14000. In PDU1 format, the bottom 8-bits are always 0x00, and the top 17 bits come from the message ID. If we look up the PGN in the .xml files, we find that this is part of the I-protocol. Specifically, it is message UO1, and contains user output values (contactor drive signals). The .xml files at this location also show the layout of the signals contained in the message.

5.2 Relationship to Mapping Configuration

Given the above message structure, it should be possible to see how this relates to how the signal mapping is configured at objects 0x3E10 to 0x3E5F in the object dictionary.

For example, when transmitting control information for say, an analogue output, the destination address part of the message ID would be populated with whatever is set in sub-index 1. Similarly, for

[1] J1939 supports two messaging formats, PDU1 and PDU2. The PDU2 format does not feature a destination address, but has support for a wider range of PGNs, so is used for broadcast data. However, for the control data that H-protocol and I-protocol, we use the PDU1 format as described as above since it supports the direct peer-to-peer addressing necessary for controlling equipment on remote devices.

[2] The only time the source address isn't populated is during the address claiming procedure, if a device is capable of performing this.

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incoming data, the mapping code checks the source address to see where the data is coming from, and unpacks that as appropriate if the source address matches the value, again, in sub-index 1.

5.3 Internal Routing

It is possible for the source address and destination address to be the same. This is common, especially for single node systems, where all signals will be effectively “routed to self”. The source address and destination address fields in the message ID will contain the same value.

This mechanism simplifies the mapping procedure somewhat, as the objects at 0x3E00 to 0x3E5F are configured in the same way, regardless of whether the data has to pass over CAN or not. If the signals are to be routed locally, they pass over an internal “virtual” CANbus. However, this process is transparent to the rest of the system.

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6 Tables of IO Signals

6.1 Digital Inputs

Signal	Handle
Drive enable	0x0200
Forward	0x0201
Reverse	0x0202
FS1	0x0203
Seat	0x0204
Handbrake	0x0205
Traction driveability profile 1 select	0x0206
Traction driveability profile 2 select	0x0207
Power steer trigger	0x0208
Inch forward	0x0209
Inch reverse	0x020A
Steer inner left	0x020B
Steer outer left	0x020C
Steer inner right	0x020D
Steer outer right	0x020E
Footbrake	0x020F
Pump 1	0x0210
Pump 2	0x0211
Pump 3	0x0212
Pump 4	0x0213
Pump 5	0x0214
Pump 6	0x0215
Pump 7	0x0216
Horn	0x0217
Pump inhibit	0x0218
Pump driveability profile 1 select	0x0219
Pump driveability profile 2 select	0x021A
Belly	0x021B
Deadman	0x021C
Odometer trip reset	0x021D
Slave left motor speed invert	0x021E
Battery electrolyte low switch	0x021F

Table 4 – Digital Input handles

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6.2 Digital Outputs

Signal	Handle
Hours counter	0x0400

Table 5 – Digital Output handles

6.3 Analogue Inputs

Signal	Handle
Traction primary throttle	0x0300
Footbrake	0x0301
Economy	0x0302
Steer	0x0303
Pump throttle 1	0x0304
Pump throttle 2	0x0305
Traction secondary throttle	0x0306

Table 6 – Analog Input handles

6.4 Analogue Outputs

Signal	Handle
Line contactor	0x0500
Pump contactor	0x0501
Power steer contactor	0x0502
Electro mechanical brake	0x0503
External LED	0x0504
Traction motor cooling fan	0x0505
Alarm buzzer	0x0506
Horn	0x0507
Brake lights	0x0508
Service indication	0x0509
Motor isolation contactor	0x050A
Precharge contactor	0x050B
High/low speed indication	0x0600

Table 7 – Analog Output handles

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